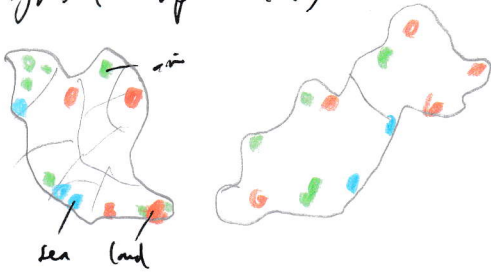
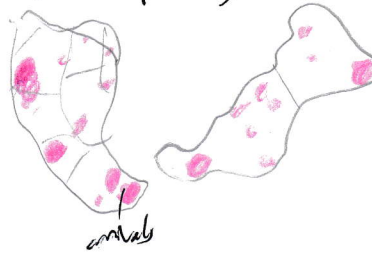


Ideas

① Symbol map (colour)



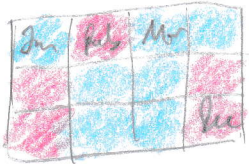
② Symbol map (size)



③ Choropleth map



④ Calendar heatmap



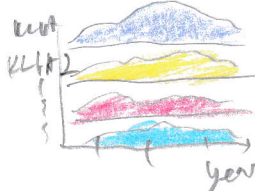
⑤ Sparklines



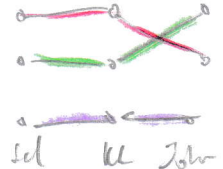
⑥ Under chart



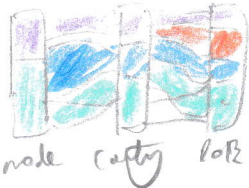
⑦ Horizon chart



⑧ Bump chart



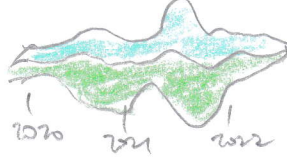
⑨ Sankey diagram



⑩ Marmelade



⑪ Stream graph



⑫ Slope chart



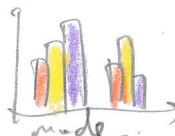
⑬ Funnel chart



⑭ Placked bar chart



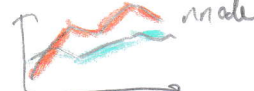
⑮ Grouped bar chart



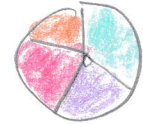
⑯ Treemap



⑰ Line chart



⑱ Pie chart



Filter

1. Slope and bump chart
similar → bump ✓ X
2. sparkline and horizon
→ sparkline ✓ X
3. pie chart too many
categories for this data X

Categorise

1. temporal patterns: ④ ⑤ ⑪ ⑰
2. composition: ① ⑤ ⑫ ⑮
3. ranking: ⑧ ⑫
4. variability: ⑥ ⑬
5. maps: ① ② ③

Combine and Refine

1. ④ + ⑰ → dual axis
bar line →
(reduce redundancy, more effort)



Questions

1. Can the visualisations clearly
show trends and patterns for
entry points?

2. Is Malaysia and Singapore
properly compared?
3. How can analysis be
easily conducted?

4. Can filtering be done
easily?

5. Does it meet the client's
(lecturer) needs?

Layout

Sheet: 2

Title: Dashboard: Monitoring Entry and Exit

Author: Bryan Leary

Date: 21 October 2025

Task: FIT3179 Assignment 2.

Overview

- Global controls (top bar).
 - State (show top few states)
 - Time window slider (2020 - 2023).
- Hover on sparkline:
 - highlights same mode for bar chart and during the other modes.
- Heatmap: hover to show value + rank in that month.
- Bar: hover to reveal per-year value, click to select in bar chart.

Discussion

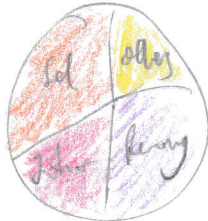
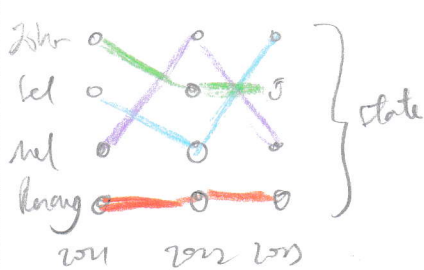
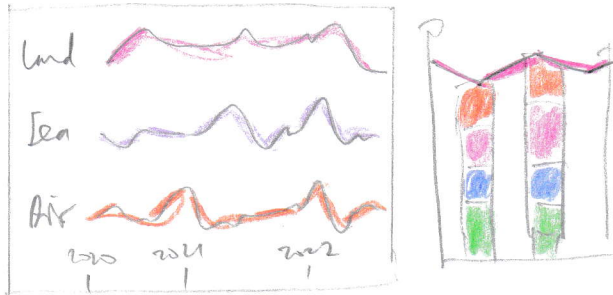
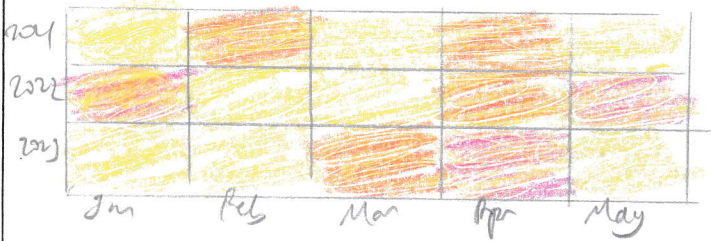
Advantages:

- single-screen shows instant seasonality. (heatmaps).
- Allow cross-state comparison, aggregated monthly, include base 2020 ??

Disadvantages:

- Has many pieces, needs strong legends and consistent colour scheme.
- Confused y-axis labels for small graphs (sparkline etc).

Entry and Exit Patterns



Focus

Primary focus: seasonality

- reveal recurring monthly patterns and how quickly each mode / state recovered after Covid.
- quick and easy comparison for analysis across time periods, eg. monthly, yearly, weekday vs weekends.

Layout

Sheet: 3

Title: Storytelling / Flow style

Author: Dym Leong

Date: 21 October 2025

Task: MTRM Assignment 2

Opening

1. Hovering over tiles is linking will show exact values, link to mode
2. Toggle state (belong, show etc) in radar chart to see different state modes.
2. Symbol map will link to survey, and highlight the same mode.
4. Same for bar chart, hovering over it will highlight the same mode in other charts.

Drawings

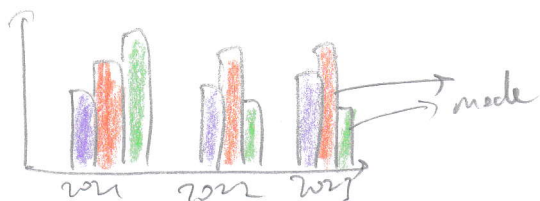
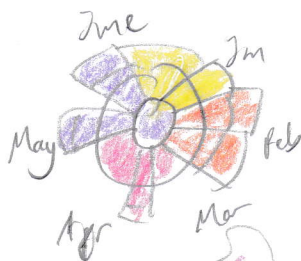
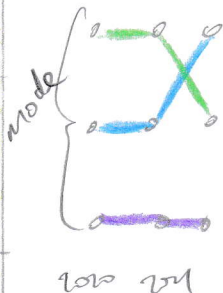
Advantages:

1. Narrative clarity. Each chart isolates a single idea, and is presented in a clear flow.
2. Clear flow in composition, good for analysis.

Disadvantages:

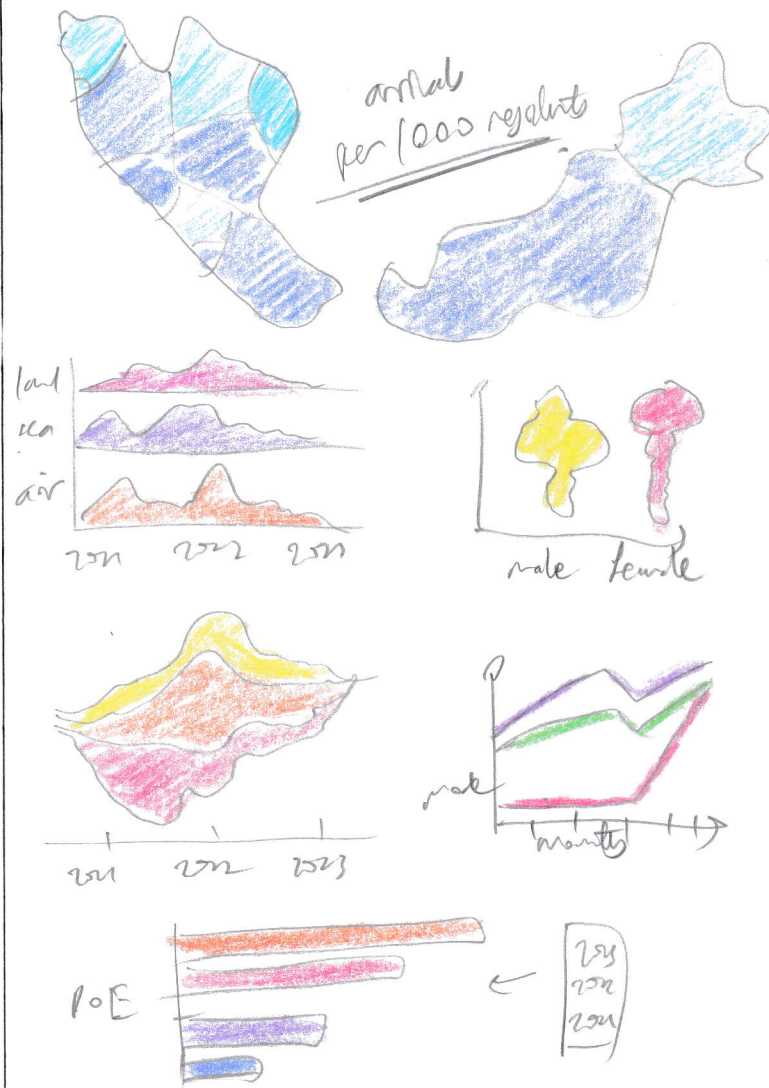
1. Category imbalance between modes (sea much less than land) may affect story and visual clarity.

Regions in Malaysia



Focus Primary focus: Composition and flow

- shows where the different modes come from, how they contribute, and how patterns change.
- clear visual hierarchy shown by map and survey, they are the main focus of the whole viz.

The Gateway of Malaysia

Title: Gateway of Malaysia

Author: Bryan Leong

Date: 21 October 2025

Task: P173129 Assignment 2

Operations

1. Puffer horizontal bar chart by year, which is also linked to line chart (yearly too).
2. Clicking a line in horizon chart will show detailed points on the male plots. Haha ??
3. Unlink chart shows details about each gender, can filter by year, or All Years.

DiscussionAdvantages:

1. Interactive metrics with clear graphs enable clear analysis.
2. Covers definitions, stability, and content show full picture.

Disadvantages:

1. Many complex charts, may be confusing without proper legends & explanation.

FocusPrimary Focus: usability and anomaly detection.

- 1st row showcases the overall picture of normalised arrivals.
- 2nd row shows distribution and splits.
- 3rd row shows stability and trends.
- 4th row shows the top few states, which link back to previous charts.

Layout

Sheet: 5

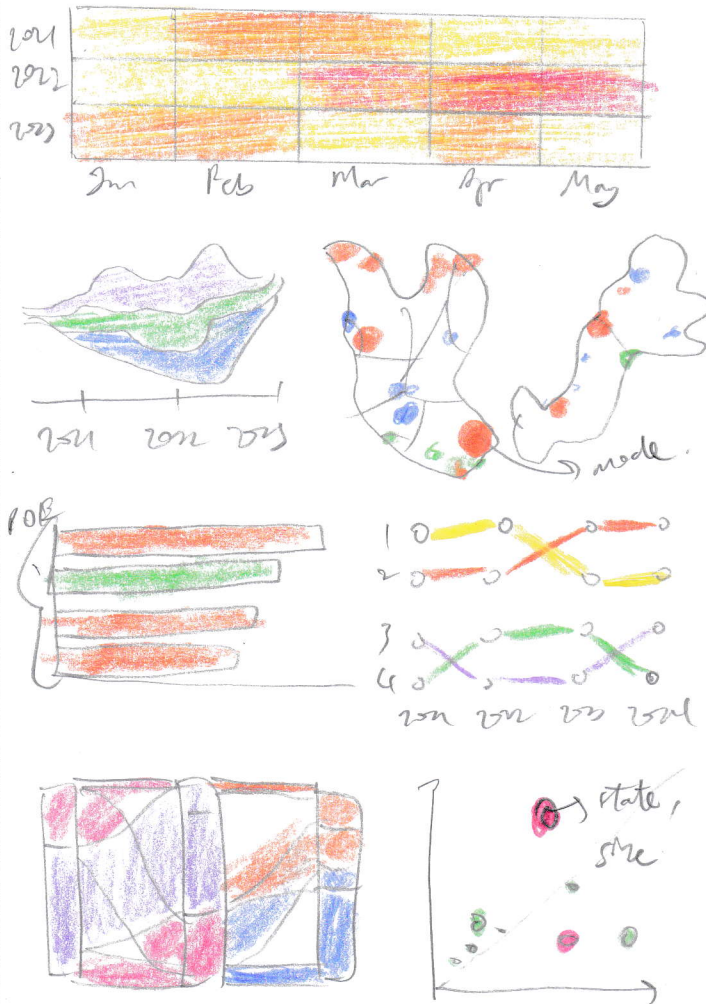
Title: The Gateway of Malaysia

Author: Bryan Leong

Date: 21 October 2025

Task: MT3129 Assignment 2.

The Gateway of Malaysia



Overview

- Global year filter (2020 - 2023).
This will affect symbol map, horizontal bar chart, and scatter plot.
- Hover on calendar heatmap or bump chart, will show / highlight the matching month in other charts.
- Bar chart can toggle to be stacked (by mode).
- Annotating for outlier on scatterplot.

Details

- Algorithm: date aggregation → by mode / state / year.
- columns: teal theme, column codes for all modes and states throughout.
- layout: small vertically, 4 distinct portions.
- dependencies: vegalite, HTML, CSS, javascript
- specific requirements:
 - no missing data (can clean)
 - hardcoded months and state names.
 - target device: small laptop, NO horizontal scroll.
- Estimated time:

$$\text{Data engineering} + \text{2 charts} + \text{code} = 8 \text{ days}$$

2 3 3

Focus: Recovery → Entry → Workers

Part 1: Recovery from covid, how the increasing country again.

Part 2: Points of entry and states of entry (distribution)

Part 3: Where the arrivals come from (country, flow: railway).

Part 4: The impact of tourism and arrivals on different states (~~scatter~~ scatterplot).